

Maia Lite 355M

Definitive Clean Pretraining Report

*Corpus Reconstruction, Deterministic Training Pipeline,
Reproducibility, and Full-Scale Pretraining*

Field	Value
Project	Maia Platform / Maia Lite
Model	Decoder-only Transformer language model
Parameters	354,823,168
Languages	English, Brazilian Portuguese, Spanish
Additional domains	Source code, mathematics, scientific literature
Primary training platform	NVIDIA A100-SXM4-80GB / Google Colab
Report date	30 September 2026
Status	Definitive clean pretraining in progress

This report documents the new clean pretraining experiment and does not replace the earlier experimental/failure-analysis record.

1. Executive Summary

Maia Lite is a compact multilingual autoregressive language model developed within the Maia Platform ecosystem. The current experiment represents a methodological reset: instead of continuing or recovering an earlier model, the model is being pretrained from randomly initialized weights using a fully materialized, deterministic, auditable corpus and a controlled training pipeline.

The definitive model contains 354,823,168 parameters, uses a 50,257-token ByteLevel BPE vocabulary, and has a 1,024-token context window. The training budget is 7,096,467,456 tokens, corresponding to 216,567 optimizer steps at 32,768 tokens per step. This yields approximately 20.000012 training tokens per model parameter, differing from the exact 20-token-per-parameter reference by only 4,096 tokens.

The canonical training corpus contains English, Brazilian Portuguese, Spanish, source-code material, mathematical material, and scientific literature. The original 849 uint16 shards are preserved for auditability, while training is performed from a single packed binary artifact copied to local SSD at runtime. A separate frozen validation corpus contains 20,480,000 tokens and 20,000 sequences.

Before training, the tokenizer, corpus identities, local I/O paths, model architecture, token budget, and dataset sizes passed a final preflight audit. The definitive run has now started on an NVIDIA A100-SXM4-80GB. Early training loss decreased from 10.283654 at step 25 to 8.534869 at step 125, with throughput near 1.00 optimizer step per second. These early measurements are encouraging but are not yet evidence of final model capability.

2. Relationship to the Earlier Experimental Program

The preceding Maia Lite experimental report documented the original multilingual Wikipedia-based pretraining, the instability observed after the 106k checkpoint, frozen-loss diagnosis, controlled recovery, deterministic corpus redesign, and later continuation experiments. That work remains part of the scientific record.

The present report begins a distinct experimental phase. The purpose is to test whether the engineering and methodological lessons learned from the earlier runs are sufficient to support a complete, reproducible pretraining trajectory from random initialization without relying on inherited model weights.

Earlier experimental program	Definitive clean pretraining
Included recovery from checkpoint 106k	Starts from random initialization
Included continuation trajectories	Single planned full-scale trajectory
Used earlier Corpus V2 variants	Uses final 7.096B-token canonical corpus
Focused partly on failure diagnosis	Focuses on reproducibility and clean pretraining
Historical checkpoints remain evidence	New checkpoints form an independent lineage

3. Research Objective

The central objective is to evaluate how far a carefully trained 355M-parameter multilingual decoder-only Transformer can be pushed when the entire pretraining process is controlled at the corpus, tokenizer, sampling, checkpointing, validation, and compute-budget levels.

The principal research question is:

How far can a compact 355M-parameter multilingual language model be pushed through reproducible pretraining, controlled corpus composition, and domain-aware post-training?

4. Model Architecture

Parameter	Value
Architecture	GPT-style decoder-only Transformer
Implementation	Hugging Face GPT2LMHeadModel
Parameters	354,823,168
Vocabulary	50,257
Context length	1,024 tokens
Layers	24
Attention heads	16
Embedding dimension	1,024
MLP inner dimension	4,096
Objective	Causal language modeling
BOS / PAD / EOS / UNK / MASK	0 / 1 / 2 / 3 / 4

The architecture is intentionally compact relative to contemporary multi-billion-parameter models. This makes it suitable for experiments targeting local inference, edge deployment, education, privacy-preserving applications, and resource-constrained environments.

5. Tokenizer Identity and Reproducibility

The definitive tokenizer is a ByteLevel BPE tokenizer with a vocabulary of 50,257 tokens. Tokenizer identity is treated as part of the corpus identity because changing tokenization would invalidate the already materialized binary corpus.

Regression text	Expected token IDs
The theory of general relativity describes gravity.	[621, 4388, 303, 1432, 42874, 9500, 14767, 18]
A inteligência artificial pode auxiliar a pesquisa científica.	[37, 15964, 7965, 1282, 19442, 261, 7151, 11490, 18]
La inteligencia artificial es un campo de la informática.	[1403, 15489, 7965, 412, 360, 4316, 281, 344, 33713, 18]
def square(x): return x * x	[9296, 7501, 12, 92, 2755, 3069, 2308, 4333, 2308]
E = mc ²	[41, 1381, 285, 71, 17471]

These regression vectors provide an exact identity test. A tokenizer that cannot reproduce these token IDs must not be used with the definitive materialized corpus.

6. Definitive Training Corpus

The definitive corpus was constructed to provide multilingual general-language coverage while reserving explicit capacity for programming, mathematics, and scientific material.

Source	Share	Training tokens
FineWeb-Edu English	45.0%	3,193,409,536
FineWeb2-HQ Portuguese	22.5%	1,596,704,768
FineWeb2-HQ Spanish	22.5%	1,596,704,768
OpenCoder FineWeb Code	5.0%	354,823,168
OpenWebMath	3.0%	212,893,696
peS2o Scientific	2.0%	141,931,520
Total	100.0%	7,096,467,456

The corpus is sequence-aligned to the 1,024-token context length and stored using unsigned 16-bit token identifiers. The canonical source representation consists of 849 shards totaling approximately 13.22 GiB. These shards are preserved rather than deleted after packing.

7. Frozen Validation Corpus

Domain	Validation tokens
FineWeb-Edu English	9,216,000
FineWeb2-HQ Portuguese	4,608,000
FineWeb2-HQ Spanish	4,608,000
OpenCoder	1,024,000
OpenWebMath	614,400
peS2o Scientific	409,600
Total	20,480,000

The validation set contains exactly 20,000 complete sequences and is disjoint from training through a deterministic selection predicate. It is frozen so that all future checkpoints can be compared against the same evaluation material.

8. Canonical Packed Artifacts

To avoid repeatedly opening hundreds of Google Drive files during training, the canonical shards are concatenated once into packed binary artifacts. The packed files remain persistent on Google Drive and are copied to the Colab local SSD when a runtime starts.

Artifact	Identity
train.bin	7,096,467,456 uint16 tokens; approximately 13.22 GiB
train.bin SHA-256	a8176dfca40f508f2fe6fd19512fcd2e78b76ab0dd4e589030f6134d0df39338
validation.bin	20,480,000 uint16 tokens; 20,000 sequences
validation.bin SHA-256	43b5584117fc52dd5e4e1068536e4acf15dd606d06c571f039001c2660b0ba70

During runtime, training reads `/content/maia_lite_v3/train.bin` and validation reads `/content/maia_lite_v3/validation.bin`. This design minimizes remote-storage I/O while retaining persistent, cryptographically identified artifacts.

9. Training Budget

Quantity	Value
Microbatch	16 sequences
Gradient accumulation	2
Effective batch	32 sequences
Tokens per optimizer step	32,768
Optimizer steps	216,567
Total training tokens	7,096,467,456
Tokens per parameter	20.000012
Exact 20× parameter reference	7,096,463,360
Difference from 20× reference	4,096 tokens

The chosen budget therefore creates a convenient experimental reference extremely close to 20 training tokens per parameter. This is an accounting reference, not an assumption that the model must stop at this point; stopping decisions will ultimately depend on validation and downstream evaluation.

10. Deterministic Sampling and Dataset Semantics

The packed training artifact is physically source-contiguous, so sequential sampling would expose the model to long source-specific blocks and would be unacceptable. The training pipeline therefore enforces global randomized sampling without replacement across the 6,930,144 training sequences.

- Global dataset size: 6,930,144 sequences.
- Sampling: randomized across the full dataset without replacement.
- Seed: 42.
- Data seed: 42.
- Deterministic sampler regression test: passed.
- Global index-range test: passed.
- Trainer data skipping on resume: enabled (`ignore_data_skip=False`).

The Trainer's outer `DataLoader` may report `SequentialSampler` because of wrapping performed by the training framework; the nested effective sampler was observed as `SeedableRandomSampler`. An explicit deterministic `RandomSampler` audit was also performed before training.

11. Training Configuration

Hyperparameter	Value
Maximum learning rate	3e-4
Warmup	2,000 optimizer steps
Scheduler	Cosine
Weight decay	0.1
Maximum gradient norm	1.0
Precision	BF16
TF32	Enabled
Microbatch	16
Gradient accumulation	2
Effective batch	32
Logging interval	25 steps
Checkpoint interval	1,000 steps
Checkpoint state	Model + optimizer + scheduler + Trainer state
Dataloader workers	2
Drop last	True
Framework	Transformers 5.17.0

The model cache is disabled during training. Full-state checkpoints are intentionally retained so that interruption and recovery can be evaluated without silently resetting optimizer or scheduler state.

12. Final Preflight Audit

Immediately before the definitive run, the complete notebook passed the following preflight checks:

Audit item	Result
Tokenizer	PASS
Training budget	PASS
Training I/O	LOCAL SSD - PASS
Validation corpus	PASS
Validation I/O	LOCAL SSD - PASS
Training read test	PASS

Validation read test	PASS
Model architecture	PASS
Packed corpus	PASS

The audited configuration reported 354,823,168 parameters, 50,257 vocabulary entries, 1,024-token context, 6,930,144 training sequences, 7,096,467,456 training tokens, 20,000 validation sequences, and 20,480,000 validation tokens.

13. Definitive Clean Pretraining: Initial Observations

The definitive run began from random initialization with no recovery checkpoint. The framework reported that an unrecognized `loss_type=None` setting would fall back to the default `ForCausalLMLoss`. This message is informational in the current run and has not been accompanied by numerical instability.

Step	Training loss	Δ from previous log	Cumulative tokens
25	10.283654	—	819,200
50	9.532371	-0.751283	1,638,400
75	9.203370	-0.329001	2,457,600
100	8.864364	-0.339006	3,276,800
125	8.534869	-0.329495	4,096,000

Between logged steps 25 and 125, training loss decreased by approximately 1.748785. At step 125, the run had processed 4,096,000 tokens, only about 0.058% of the planned training budget. The observed throughput has been approximately 1.00 optimizer step per second, implying a gross runtime near 60 hours if sustained.

The early loss trajectory is consistent with successful optimization, but it is too early to draw conclusions about final generalization, multilingual balance, domain capability, or useful generation quality.

14. Checkpoint and Recovery Protocol

Recovery is a first-class experimental requirement because the execution environment may be interrupted by session limits, compute-credit exhaustion, or runtime replacement.

1. Save a full checkpoint every 1,000 optimizer steps.
2. Require `trainer_state.json`, `optimizer.pt`, and `scheduler.pt` for a checkpoint to qualify as a full recovery checkpoint.
3. Resume through `Trainer.train(resume_from_checkpoint=...)` rather than rebuilding optimizer and scheduler state manually.
4. Preserve `ignore_data_skip=False` so that the framework can advance through the deterministic data order on resume.
5. At the first suitable checkpoint, perform a controlled recovery audit and compare post-resume behavior with the uninterrupted trajectory.

At the time of this report, the clean run has not yet reached its first checkpoint. Consequently, exact full-state resume behavior for this new run remains to be empirically demonstrated rather than assumed.

15. Planned Evaluation

The definitive model will not be judged from training loss alone. The frozen validation corpus and downstream evaluations are required to characterize the result.

- Global frozen-validation loss and perplexity.
- English, Portuguese, and Spanish loss/perplexity.
- Code-domain loss/perplexity.
- Mathematical-domain loss/perplexity.
- Scientific-text loss/perplexity.
- Natural base-model continuation in all three languages.
- Scientific, mathematical, and programming continuations.
- Standardized external benchmarks where methodologically appropriate.
- Post-SFT instruction-following evaluation.
- Quantized inference and edge-deployment measurements.

16. Planned Supervised Fine-Tuning

After pretraining, Maia Lite is expected to undergo a unified supervised fine-tuning stage rather than isolated sequential specialization. The planned principal axes are multilingual conversation, Python/programming, and computational-modeling/scientific material.

Earlier SFT experiments belong to the historical program and must not be interpreted as evaluations of the new clean base model. The SFT dataset and pipeline will therefore be reconstructed and audited before they are applied to the definitive pretrained checkpoint.

17. Reproducibility Assets

- Frozen tokenizer files and exact token-ID regression vectors.
- Corpus-construction scripts and source manifests.
- 849 canonical training shards.
- Packed train.bin and validation.bin artifacts.
- SHA-256 identities for packed artifacts.
- Frozen validation corpus.
- Notebook and training configuration.
- Model architecture specification.
- Random seeds and sampling audit.
- Checkpoints including optimizer, scheduler, and Trainer state.

- Training logs and validation outputs.
- Hardware, CUDA, PyTorch, Transformers, and environment information.
- Compute consumption records.

Upstream dataset licenses and redistribution terms must be reviewed individually before any derived corpus is redistributed. Where redistribution is restricted, reconstruction scripts, source identifiers, manifests, filtering rules, and hashes should be preserved instead.

18. Methodological Safeguards

- Do not modify the tokenizer after corpus materialization.
- Do not replace canonical corpus artifacts without changing their recorded identity.
- Do not infer model quality from training loss alone.
- Do not conflate historical recovery checkpoints with the new clean checkpoint lineage.
- Do not claim exact resume reproducibility until it has been tested on the clean run.
- Do not alter validation data between checkpoints.
- Preserve unsuccessful experiments and anomalies as part of the scientific record.
- Distinguish observations, hypotheses, interventions, and interpretations in future revisions.

19. Current Status

- Definitive tokenizer identity: verified.
- Canonical 7.096B-token training corpus: complete.
- 849 original shards: preserved.
- Packed training artifact: verified.
- Frozen 20.48M-token validation artifact: verified.
- Model architecture: verified at exactly 354,823,168 parameters.
- Training budget: verified at exactly 216,567 optimizer steps.
- Global randomized sampling audit: passed.
- Final preflight: passed.
- Definitive clean pretraining: running on NVIDIA A100-SXM4-80GB.
- Latest observation included in this report: step 125, training loss 8.534869.

The immediate milestone is checkpoint 1,000, followed by verification of checkpoint completeness and, when operationally convenient, a controlled full-state resume test. The warmup ends at optimizer step 2,000. The long-term milestone is completion and frozen evaluation of step 216,567.

20. Preliminary Interpretation

The new experiment has successfully passed the transition from corpus construction and pipeline auditing to actual optimization. The first logged losses show rapid early learning from random initialization, while throughput is stable and no numerical failure has appeared.

The strongest methodological result at this stage is not model quality but experimental control: the corpus size, model size, tokenizer identity, sampling strategy, training budget, validation set, and binary artifacts are all explicitly defined and auditable.

No claim about final Maia Lite capability is justified yet. Such conclusions require completion of the planned trajectory and controlled evaluation of the resulting checkpoints.

21. Next Experimental Milestones

Milestone	Purpose
Step 1,000	First full checkpoint; verify recovery-state files
Step 2,000	End of learning-rate warmup
Intermediate checkpoints	Monitor loss stability and preserve evidence
Step 216,567	Complete planned 7.096B-token clean pretraining budget
Frozen validation	Measure global and domain-specific generalization
Base generation tests	Assess pretrained generative structure before SFT
Unified SFT	Conversation + programming + computational/scientific material
Quantization/edge tests	Characterize quality-memory-latency trade-offs

Appendix A. Definitive Numeric Reference

Reference	Value
Parameter count	354,823,168
Vocabulary	50,257
Context	1,024
Training sequences	6,930,144
Training tokens	7,096,467,456
Validation sequences	20,000
Validation tokens	20,480,000
Effective batch	32 sequences
Tokens/step	32,768
Optimizer steps	216,567
Tokens/parameter	20.000012
train.bin SHA-256	a8176dfca40f508f2fe6fd19512fcd2e78b76ab0dd4e589030f6134d0df39338
validation.bin SHA-256	43b5584117fc52dd5e4e1068536e4acf15dd606d06c571f039001c2660b0ba70

Report status: Living experimental document

Next revision: after the first checkpoint/recovery audit and subsequent training milestones.